SAST – TECH BRANCH DOCUMENTATION

SECTION 1: BRANCH OVERVIEW

The Tech Branch of SAST is dedicated to advancing the integration of Artificial Intelligence (AI), Machine Learning (ML), and software engineering within the space technology domain. Its mission is to position SAST as a leader in software innovation for SpaceTech by designing, developing, and deploying intelligent systems that power data-driven insights, automation, and mission efficiency.

The branch plays a vital role in collaboration with the Product Branch, helping transform product ideas into fully functional, high-performance systems. This includes backend development, AI/ML modules, data visualization, and workflow automation to deliver a seamless technical user experience (UX).

Through innovation, research, and collaboration, the Tech Branch aims to expand the use of technology in all SAST projects empowering the club and the university ecosystem to explore the future of space through computation and creativity.

SECTION 2: CORE FOCUS AREAS

AI and ML for Analysis and Decision-Making
Leveraging artificial intelligence and machine learning to analyze satellite, mission, and sensor data enabling automated insights, anomaly detection, and predictive modeling in space research contexts.

Web and App Development for Space-Tech Solutions
Designing and developing scalable digital platforms that support SAST's operations, research projects, and public engagement from mission management dashboards to educational and simulation tools.

Data Visualization and Interpretation
Transforming complex datasets into intuitive, interactive visualizations that help teams understand patterns, outcomes, and scientific conclusions effectively.

Automation Systems and Workflow Optimization
Building automated tools and systems to streamline repetitive or time-consuming technical

processes, ensuring efficiency in data processing, analysis, and mission operations.

SECTION 3: ASSIGNMENT OVERVIEW

Task:
As part of the recruitment and ideation process, members are required to submit a Technical Proposal Document (TPD) that presents:
- How technology (AI, ML, Web/App Development, or Automation) can increase involvement and efficiency in SAST projects.
- A concept or prototype such as a web application, visualization tool, or automation script built around space-tech concepts using real or mock data.
- An idea or roadmap for future technical projects that the Tech Branch or SAST could pursue.

Goal:
The objective is to demonstrate technical creativity, practical understanding, and vision for expanding the role of software and AI in the space-tech ecosystem of Rishihood University.

SECTION 4: TECHNICAL PROPOSAL FORMAT (MANDATORY SECTIONS)

1. Technical Overview

2. Problem Definition

3. Target Users / Applications

4. Proposed Solution

5. System Design / Architecture

6. Tech Stack

7. Implementation Plan

8. Feasibility and Challenges

9. Success Metrics

10. Timeline and Milestones

SECTION 5: EVALUATION CRITERIA

Criteria and Weightage:

Innovation and Technical Creativity	25%
Problem Understanding	15%
System Feasibility and Implementation Depth	20%
Practical Impact for SAST	20%
Documentation Clarity and Presentation	10%
SAST Alignment (Relevance to Space/Tech)	10%

SECTION 6: OUTCOME AND OPPORTUNITIES

Top submissions will:

- Be considered for incubation and development under the Tech Branch, with mentorship and technical guidance from SAST.
- Earn an opportunity to lead or co-develop software systems for real SAST missions and research projects.
- Receive mentorship from experts in AI, ML, and software engineering within and outside the university.
- Collaborate directly with the Product Branch to translate product concepts into deployable technical systems.
- Be considered for positions such as Tech Lead, Developer, or Research Engineer within SAST.